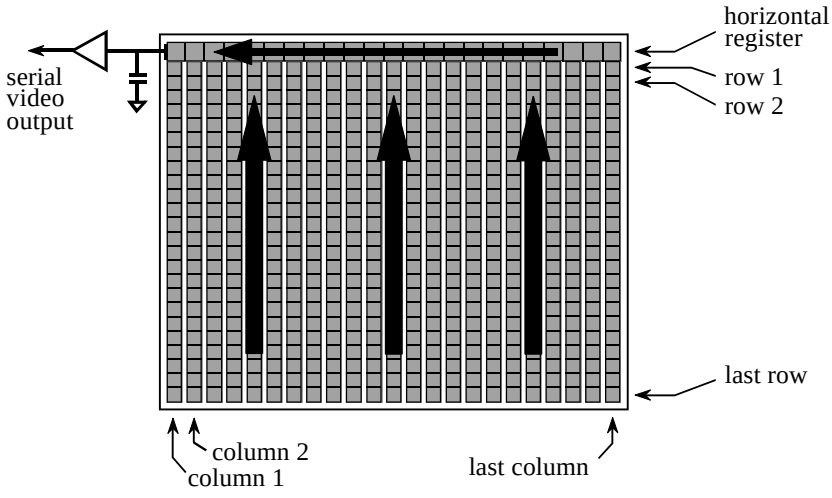

FIGURE 23-8
Architecture of the CCD. The imaging wells of the CCD are arranged in columns. During readout, the charge from each well is moved up the column into a horizontal register. The horizontal register is then readout into the charge sensitive preamplifier.

Notice that this architecture converts a two-dimensional array into a serial data stream in a particular sequence. The first pixel to be read is at the top-left corner of the image. The readout then proceeds from left-to-right on the first line, and then continues from left-to-right on subsequent lines. This is called **row major order**, and is almost always followed when a two-dimensional array (image) is converted to sequential data.

Television Video Signals

Although over 50 years old, the standard television signal is still one of the most common way to transmit an image. Figure 23-9 shows how the television signal appears on an oscilloscope. This is called **composite video**, meaning that there are vertical and horizontal synchronization (sync) pulses mixed with the actual picture information. These pulses are used in the television receiver to synchronize the vertical and horizontal deflection circuits to match the video being displayed. Each second of standard video contains 30 complete images, commonly called **frames**. A video engineer would say that each frame contains 525 **lines**, the television jargon for what programmers call *rows*. This number is a little deceptive because only 480 to 486 of these lines contain video information; the remaining 39 to 45 lines are reserved for sync pulses to keep the television's circuits synchronized with the video signal.

Standard television uses an **interlaced** format to reduce *flicker* in the displayed image. This means that all the odd lines of each frame are transmitted first, followed by the even lines. The group of odd lines is called the **odd field**, and the group of even lines is called the **even field**. Since